



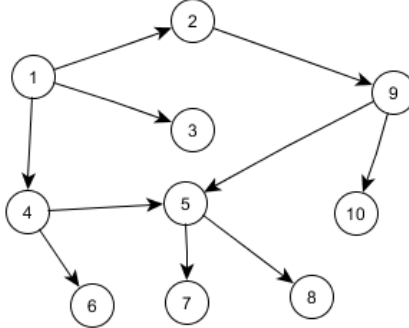
NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008

Department	Computer Science and Engineering
Examination	End Semester – 2024-25
Program	B. Tech / M. Tech / Ph. D
Session	Spring
Course code & Course Name	CS6404 - Software Testing
Marks	50
Duration	3 Hours

This question paper contains 2 pages

Answer all questions.

Q. No.	Questions	Marks	CO*	BL**
1.	Consider the following interest calculation problem: If depositing less than Rs. 1 Lakh, rate of interest is calculated as follows: 6% for deposit upto 1 year; 7% for deposit over 1 year but less than 3 years; 8% for deposit 3 years and above. If depositing more than Rs.1Lakh, rate of interest is calculated as follows: 7% for deposit upto 1 year, 8% for deposit over 1 year but less than 3 years, 9% for deposit 3 years and above. Design test cases for the above interest calculation problem using cause-effect graph based testing technique.	5	CO-1	BL-3
2.	Consider an application <i>NameApp</i> that takes two inputs <i>name</i> and <i>age</i> where <i>name</i> is a non-empty string containing at most 25 alphabetic characters and <i>age</i> is an integer that must satisfy the constraint $0 \leq \text{age} \leq 100$. Design test cases using worst-case boundary value analysis method.	5	CO-1	BL-3
3.	Consider the following program segment: <pre>main() { int number, index; printf("Enter the number"); scanf("%d", &number); index=2; while(index<=number - 1) { if(number % index ==0) { printf("Not a prime number"); break; } index++; } if(index==number) printf("Prime number"); }</pre> Draw the CFG for the above function. Determine the cyclomatic complexity. List all the independent paths. Design test cases for all independent paths.	5	CO-2	BL-3

4.	Consider the program given below for calculating the gross salary of an employee in an organization. <pre>main() { 1. float bs, gs, da, hra = 0; 2. printf("Enter basic salary"); 3. scanf("%f", &bs); 4. if(bs < 1500) 5. { 6. hra = bs * 10/100; 7. da = bs * 90/100; 8. } 9. else 10. { 11. hra = 500; 12. da = bs * 98/100; 13. } 14. gs = bs + hra + da; 15. printf("Gross Salary = Rs. %f", gs); 16. }</pre> Find out the define-use-kill patterns for all the variables in this program.	5	CO-2	BL-3																								
5.	Suppose you have to develop a software to automate the activities of a large library. Explain the different types of performance (non-functional) testing that you will perform for the above software. You may assume your own functional and non-functional requirements.	5	CO-3	BL-3																								
6.	Perform neighborhood integration testing for the call graph shown below and calculate the effort required for this testing. 	5	CO-3	BL-3																								
7.	What do you mean by regression test prioritization? Explain the regression test prioritization technique along with a suitable diagram and the necessary steps.	5	CO-4	BL-2																								
8.	<table><tr><th>Sl. No.</th><th>Functionality</th><th>Function ID in SRS</th><th>Test Cases</th></tr><tr><td>1</td><td>Login the System</td><td>F3.4</td><td>T1</td></tr><tr><td>2</td><td>View Inventory Details</td><td>F3.5</td><td>T2</td></tr><tr><td>3</td><td>Order Book</td><td>F3.6</td><td>T3</td></tr><tr><td>4</td><td>Manage Inventory</td><td>F3.7</td><td>T4</td></tr><tr><td>5</td><td>Exit the System</td><td>F3.8</td><td>T5</td></tr></table> Develop the test specification & test log for ‘View Inventory Details’ functionality.	Sl. No.	Functionality	Function ID in SRS	Test Cases	1	Login the System	F3.4	T1	2	View Inventory Details	F3.5	T2	3	Order Book	F3.6	T3	4	Manage Inventory	F3.7	T4	5	Exit the System	F3.8	T5	5	CO-4	BL-4
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9.	Apply Modal Class Test Pattern to <i>DebitCard</i> class and generate the test cases. OR What is partition class testing? What are the different types of partition class testing? Generate 2 testcases for the <i>DebitCard</i> class using each type of partition class testing. You may assume your own operations / methods for the <i>DebitCard</i> class.	5	CO-5	BL-4																								
10.	Explain the agile testing life cycle with a suitable diagram. OR Suppose you are given the task of developing a web application for student admission in NIT Rourkela. Briefly explain the following types of web testing that you need to perform for the above web application. (i) Interface testing (ii) Load testing	5	CO-5	BL-2																								